

GradeSense - Candidate Answer Sheet

Name: Aarav Menon

Roll No: X-B / 24

Total marks: 15

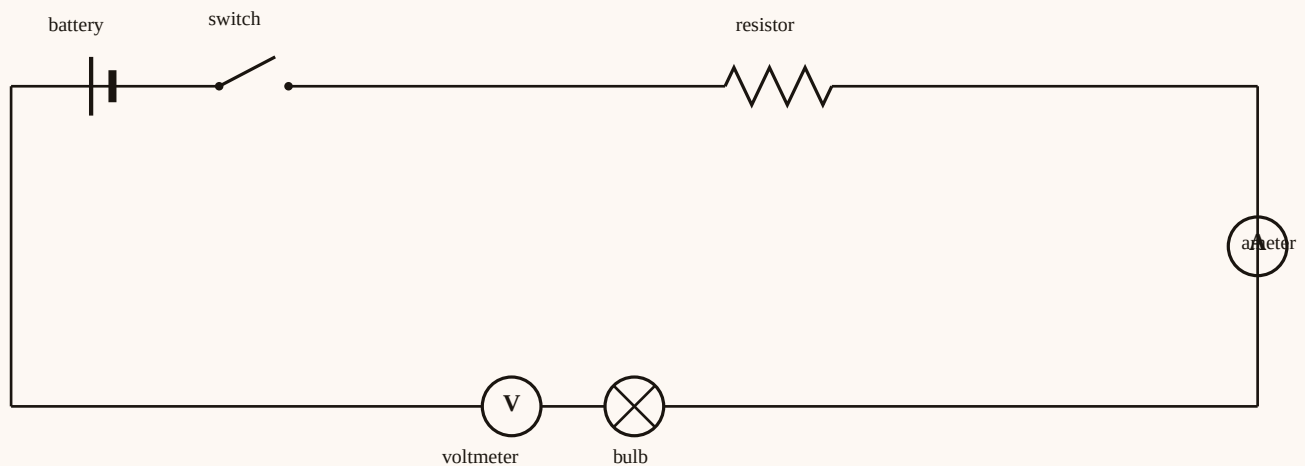
Answer 1 - Science

An electric circuit is a closed path which allows the electric current to flow through it. The battery is the source which gives the potential difference to the circuit and pushes the current forward. The switch is used to make the circuit open or close. When the switch is closed the path is complete and the current flows through the bulb and the resistor, and when the switch is open the path is broken so no current can flow.

In my diagram the battery, the switch, the resistor, the bulb and the ammeter are all joined one after another in series in the main circuit. The ammeter is connected in series because it has to measure the current which is passing through the circuit. The voltmeter is also connected in series with the bulb so that it can measure the voltage across the bulb.

The current in the circuit depends on the resistance. Ohm's law is $V = I/R$. If the resistance is increased and the battery voltage stays the same then less current will flow through the circuit, and if we decrease the resistance then more current will flow.

Circuit diagram



Answer 2 - English

In my opinion technology has made students very dependent on easy answers and it does not make them better learners. Earlier a student had to go to the library and search through many different books, and that whole process of searching made them think deeply about the topic. Now a student simply types the question into their phone and the answer appears in one second, so their brain does not have to do any work at all.

Students are now completely used to getting everything instantly and a lot of them cannot solve even a simple problem without looking it up first. Everybody knows that students just copy from the internet and do not learn anything, this is happening in every single school and in every single class and it is a very big problem which is making students weaker and weaker every year and the teachers all know about it also.

Some people say that technology is helpful.

So in conclusion I believe that technology is making students dependent on ready-made answers instead of making them into better learners, and students should always try to solve a problem by themselves first before they search for the answer.

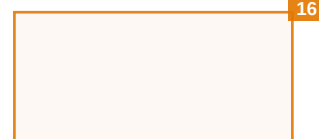
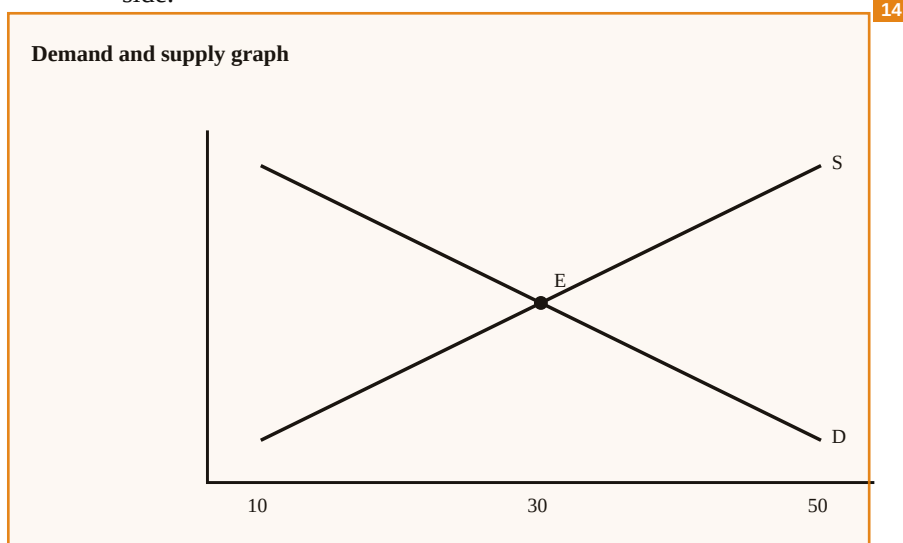
Answer 3 - Economics

From the table we can see that when the price increases the quantity demanded goes down and at the same time the quantity supplied goes up. I have drawn the graph below. The demand curve slopes downward and the supply curve slopes upward.

The two curves cut each other at the price of Rs 30 and the quantity of 60 units. This is the equilibrium point of the market because at this price the quantity demanded is exactly equal to the quantity supplied, so there is no pressure on the price to change.

When the price is below the equilibrium price there is a surplus in the market, because the producers are supplying much more than what the consumers actually want to buy. When the price is above the equilibrium price there is a shortage in the market.

If the cost of production increases then the producers will not be able to supply the same quantity at the same price as before, so they will supply less at every price. Therefore the supply curve will shift towards the left side.



Marking summary

7.5 out of 15 marks confidence 73% marked by mock/rule-based-mock

This paper needs a human check before the marks are used:

Question 3 was marked with only 61% confidence.

Question 1 - Science: 2.5 / 5

Question 1: 2.5 of 5 marks. 2 rubric points fully met. 1 contains a substantive error. 1 not addressed.

1/1 Correctly represents the main circuit with battery, switch, bulb and resistor connected in a closed series path

The main circuit is correctly described as a closed series loop containing the battery, switch, resistor and bulb.

Evidence: "An electric circuit is a closed path which allows the electric current to flow through it."

0/1 Correct placement of ammeter in series and voltmeter in parallel across the bulb

A voltmeter must be connected in parallel across the bulb, not in series with it. In series it would carry the circuit current and could not measure the potential difference across the bulb. The diagram repeats this error.

Evidence: "The voltmeter is also connected in series with the bulb so that it can measure the voltage across the bulb."

Correction: Connect the voltmeter in parallel across the bulb, so it measures the potential difference between the bulb's two ends.

1/1 Correct explanation of current flow and the function of the main components

The functions of the battery and switch, and the path the current follows when the switch is closed, are correctly explained.

Evidence: "The battery is the source which gives the potential difference to the circuit and pushes the current forward."

0.5/1 Correctly explains the relationship between resistance and current, including the relevant principle/Ohm's law

The reasoning is right - increasing resistance does reduce the current - but Ohm's law is written incorrectly. $V = I/R$ would mean voltage falls as current rises.

Evidence: "battery voltage stays the same then less current will flow through the circuit, and if we decrease the resistance"

Correction: Ohm's law is $V = IR$, which rearranges to $I = V / R$.

0/1 Clear, logically structured explanation with appropriate labels and current direction in the diagram

The diagram does not show the conventional current direction, and the answer never mentions it. The diagram also runs past the right-hand margin, with the ammeter label overlapping the wire.

Correction: Mark an arrow showing conventional current flowing from the positive terminal of the battery around the external circuit, and keep the diagram and its labels inside the margins.

Question 2 - English: 3 / 5

Question 2: 3 of 5 marks. 3 rubric points fully met. 2 contain a substantive error.

1/1 Presents a clear position on whether/how technology affects learning

A clear position on the effect of technology on learning is stated up front.

Evidence: "In my opinion technology has made students very dependent on easy answers and it does not make them better"

1/1 Provides relevant and logically developed arguments supporting the position

The position is supported with a developed line of reasoning rather than assertion alone.

Evidence: "Earlier a student had to go to the library and search through many different books, and that whole process"

0/1 Recognises and meaningfully addresses an opposing viewpoint or limitation

The opposing viewpoint is mentioned in a single sentence and then dropped. To earn this mark it has to be engaged with - what is the strongest version of that view, and why do you still disagree?

Evidence: "Some people say that technology is helpful."

Correction: Expand this into two or three sentences: acknowledge that technology genuinely helps students who use it to understand rather than to copy, then explain why you think dependence still outweighs that benefit.

0/1 Uses relevant examples and demonstrates reasoning rather than merely making unsupported claims

This is an unsupported sweeping claim rather than evidence. "Everybody knows" and "every single school" assert what the argument needs to demonstrate, and no concrete example is given.

Evidence: "Everybody knows that students just copy from the internet and do not learn anything, this"

Correction: Replace this with a specific example, such as a student who copies a worked solution, passes the homework, and then cannot attempt a similar problem in a test.

1/1 Provides a coherent conclusion that follows from the discussion, with clear overall communication

The conclusion follows from the argument that precedes it and the answer reads clearly throughout.

Evidence: "So in conclusion I believe that technology is making students dependent on ready made answers instead of making"

Question 3 - Economics: 2 / 5

Question 3: 2 of 5 marks. 2 rubric points fully met. 1 contains a substantive error. 2 not addressed.

0/1 Correctly plots and labels the demand and supply curves, with appropriate axes and direction

The graph has price on the horizontal axis and quantity on the vertical axis, which is the wrong way round, and neither axis is labelled.

Correction: Redraw with quantity on the horizontal axis and price on the vertical axis, and label both axes including the units.

1/1 Correctly identifies the equilibrium at Rs 30 and 60 units and explains why it is equilibrium

The equilibrium is correctly identified at Rs 30 and 60 units, with the reason that quantity demanded equals quantity supplied.
Evidence: "The two curves cut each other at the price of Rs 30 and the quantity of 60 units."

0/1 Correctly explains shortage below equilibrium and surplus above equilibrium

Shortage and surplus are the wrong way round. Below equilibrium the price is low, so buyers want more than producers will supply - that is a shortage. A surplus happens above equilibrium.

Evidence: "When the price is below the equilibrium price there is a surplus in the market, because the producers are supplying"

Correction: Below equilibrium, quantity demanded exceeds quantity supplied, creating a shortage that pushes the price up. Above equilibrium, quantity supplied exceeds quantity demanded, creating a surplus that pushes the price down.

1/1 Correctly explains that increased production costs shift the supply curve left/upward

A rise in production costs is correctly shown to shift the supply curve left.

Evidence: "Therefore the supply curve will shift towards the left"

0/1 Correctly explains the resulting tendency toward a higher equilibrium price and lower equilibrium quantity, with the change represented appropriately on the graph

The answer stops at the supply curve shifting left and never says what happens to the equilibrium price and quantity as a result.

Correction: Add that the new supply curve meets unchanged demand at a higher equilibrium price and a lower equilibrium quantity, and show the new intersection on the graph.

Annotations on the paper

1. [Incorrect] page 1

A voltmeter must be connected in parallel across the bulb, not in series with it. In series it would carry the circuit current and could not measure the potential difference across the bulb. The diagram repeats this error.

Connect the voltmeter in parallel across the bulb, so it measures the potential difference between the bulb's two ends.

2. [Incorrect] page 1

The reasoning is right - increasing resistance does reduce the current - but Ohm's law is written incorrectly. $V = I/R$ would mean voltage falls as current rises.

Ohm's law is $V = IR$, which rearranges to $I = V / R$.

3. [Missing] page 1

The diagram does not show the conventional current direction, and the answer never mentions it. The diagram also runs past the right-hand margin, with the ammeter label overlapping the wire.

Mark an arrow showing conventional current flowing from the positive terminal of the battery around the external circuit, and keep the diagram and its labels inside the margins.

4. [Spelling] page 1

"circuit" should be "circuit".

circuit

5. [Spelling] page 1

"potencial" should be "potential".

potential

6. [Spelling] page 1

"ameter" should be "ammeter".

ammeter

7. [Spelling] page 1

"resistence" should be "resistance".

resistance

8. [Incorrect] page 1

The opposing viewpoint is mentioned in a single sentence and then dropped. To earn this mark it has to be engaged with - what is the strongest version of that view, and why do you still disagree?

Expand this into two or three sentences: acknowledge that technology genuinely helps students who use it to understand rather than to copy, then explain why you think dependence still outweighs that benefit.

9. [Incorrect] page 1

This is an unsupported sweeping claim rather than evidence. "Everybody knows" and "every single school" assert what the argument needs to demonstrate, and no concrete example is given.

Replace this with a specific example, such as a student who copies a worked solution, passes the homework, and then cannot attempt a similar problem in a test.

10. [Spelling] page 1

"alot" should be "a lot".

a lot

11. [Grammar] page 1

"there brain" should be "their brain".

their brain

12. [Grammar] page 1

"there phone" should be "their phone".

their phone

13. [Grammar] page 1

"Students is" should be "Students are".

Students are

14. [Missing] page 2

The graph has price on the horizontal axis and quantity on the vertical axis, which is the wrong way round, and neither axis is labelled. Redraw with quantity on the horizontal axis and price on the vertical axis, and label both axes including the units.

15. [Incorrect] page 2

Shortage and surplus are the wrong way round. Below equilibrium the price is low, so buyers want more than producers will supply - that is a shortage. A surplus happens above equilibrium.

Below equilibrium, quantity demanded exceeds quantity supplied, creating a shortage that pushes the price up. Above equilibrium, quantity supplied exceeds quantity demanded, creating a surplus that pushes the price down.

16. [Missing] page 2 (margin note - could not be placed)

The answer stops at the supply curve shifting left and never says what happens to the equilibrium price and quantity as a result.

Add that the new supply curve meets unchanged demand at a higher equilibrium price and a lower equilibrium quantity, and show the new intersection on the graph.